

## Chapter 5: Locus and Straight Line

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### EXERCISE 5.1 [PAGE 67]

#### Exercise 5.1 | Q 1 | Page 67

If A(1, 3) and B(2, 1) are points, find the equation of the locus of point P such that  $PA = PB$ .

#### SOLUTION

Let P(x, y) be any point on the required locus.

Given, A(1, 3) and B(2, 1).

$$PA = PB$$

$$\therefore PA^2 = PB^2$$

$$\therefore (x - 1)^2 + (y - 3)^2 = (x - 2)^2 + (y - 1)^2$$

$$\therefore x^2 - 2x + 1 + y^2 - 6y + 9 = x^2 - 4x + 4 + y^2 - 2y + 1$$

$$\therefore -2x - 6y + 10 = -4x - 2y + 5$$

$$\therefore 2x - 4y + 5 = 0$$

$$\therefore \text{The required equation of locus is } 2x - 4y + 5 = 0.$$

#### Exercise 5.1 | Q 2 | Page 67

A(-5, 2) and B(4, 1). Find the equation of the locus of point P, which is equidistant from A and B.

#### SOLUTION

Let P(x, y) be any point on the required locus.

P is equidistant from A(-5, 2) and B(4, 1).

$$\therefore PA = PB$$

$$\therefore PA^2 = PB^2$$

$$\therefore (x + 5)^2 + (y - 2)^2 = (x - 4)^2 + (y - 1)^2$$

$$\therefore x^2 + 10x + 25 + y^2 - 4y + 4 = x^2 - 8x + 16 + y^2 - 2y + 1$$

$$\therefore 10x - 4y + 29 = -8x - 2y + 17$$

$$\therefore 18x - 2y + 12 = 0$$

$$\therefore 9x - y + 6 = 0$$

$$\therefore \text{The required equation of locus is } 9x - y + 6 = 0.$$

#### Exercise 5.1 | Q 3 | Page 67

If A(2, 0) and B(0, 3) are two points, find the equation of the locus of point P such that  $AP = 2BP$ .

**SOLUTION**

Let  $P(x, y)$  be any point on the required locus.

Given,  $A(2, 0)$ ,  $B(0, 3)$  and

$$AP = 2BP$$

$$\therefore AP^2 = 4BP^2$$

$$\therefore (x - 2)^2 + (y - 0)^2 = 4[(x - 0)^2 + (y - 3)^2]$$

$$\therefore x^2 - 4x + 4 + y^2 = 4(x^2 + y^2 - 6y + 9)$$

$$\therefore x^2 - 4x + 4 + y^2 = 4x^2 + 4y^2 - 24y + 36$$

$$\therefore 3x^2 + 3y^2 + 4x - 24y + 32 = 0$$

$\therefore$  The required equation of locus is

$$3x^2 + 3y^2 + 4x - 24y + 32 = 0$$

**Exercise 5.1 | Q 4 | Page 67**

If  $A(4, 1)$  and  $B(5, 4)$ , find the equation of the locus of point  $P$  if  $PA^2 = 3PB^2$ .

**SOLUTION**

Let  $P(x, y)$  be any point on the required locus.

Given,  $A(4, 1)$ ,  $B(5, 4)$  and

$$PA^2 = 3PB^2$$

$$\therefore (x - 4)^2 + (y - 1)^2 = 3[(x - 5)^2 + (y - 4)^2]$$

$$\therefore x^2 - 8x + 16 + y^2 - 2y + 1 = 3(x^2 - 10x + 25 + y^2 - 8y + 16)$$

$$\therefore x^2 - 8x + y^2 - 2y + 17 = 3x^2 - 30x + 75 + 3y^2 - 24y + 48$$

$$\therefore 2x^2 + 2y^2 - 22x - 22y + 106 = 0$$

$$\therefore x^2 + y^2 - 11x - 11y + 53 = 0$$

$\therefore$  The required equation of locus is

$$x^2 + y^2 - 11x - 11y + 53 = 0.$$

**Exercise 5.1 | Q 5 | Page 67**

$A(2, 4)$  and  $B(5, 8)$ , find the equation of the locus of point  $P$  such that  $PA^2 - PB^2 = 13$ .

**SOLUTION**

Let  $P(x, y)$  be any point on the required locus.

Given,  $A(2, 4)$ ,  $B(5, 8)$  and

$$PA^2 - PB^2 = 13$$

$$\therefore [(x - 2)^2 + (y - 4)^2] - [(x - 5)^2 + (y - 8)^2] = 13$$

$$\therefore (x^2 - 4x + 4 + y^2 - 8y + 16) - (x^2 - 10x + 25 + y^2 - 16y + 64) = 13$$

$$\therefore 6x + 8y - 69 = 13$$

$$\therefore 6x + 8y - 82 = 0$$

$$\therefore 3x + 4y - 41 = 0$$

$\therefore$  The required equation of locus is

$$3x + 4y - 41 = 0.$$

### Exercise 5.1 | Q 6 | Page 67

A(1, 6) and B(3, 5), find the equation of the locus of point P such that segment AB subtends right angle at P. ( $\angle APB = 90^\circ$ )

#### SOLUTION

Let P(x, y) be any point on the required locus.

Given, A(1, 6) and B(3, 5),

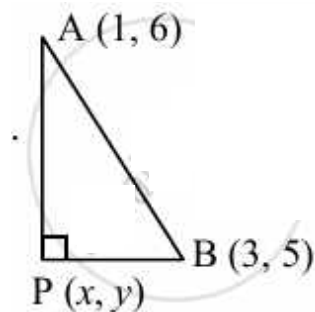
( $\angle APB = 90^\circ$ )

$\therefore \angle APB = 90^\circ$

$\therefore \triangle APB$  is a right angled triangle.

By Pythagoras theorem,

$$AP^2 + PB^2 = AB^2$$



$$\begin{aligned}\therefore [(x-1)^2 + (y-6)^2] + [(x-3)^2 + (y-5)^2] &= (1-3)^2 + (6-5)^2 \\ \therefore x^2 - 2x + 1 + y^2 - 12y + 36 + x^2 - 6x + 9 + y^2 - 10y + 25 &= 4 + 1 \\ \therefore 2x^2 + 2y^2 - 8x - 22y + 66 &= 0 \\ \therefore x^2 + y^2 - 4x - 11y + 33 &= 0 \\ \therefore \text{The required equation of locus is} \\ x^2 + y^2 - 4x - 11y + 33 &= 0.\end{aligned}$$

### Exercise 5.1 | Q 7.1 | Page 67

If the origin is shifted to the point O'(2, 3), the axes remaining parallel to the original axes, find the new co-ordinates of the points A(1, 3)

#### SOLUTION

Origin is shifted to (2, 3) = (h, k)

Let the new co-ordinates be (X, Y).

$$\therefore x = X + h \text{ and } y = Y + k$$

$$\therefore x = X + 2 \text{ and } y = Y + 3 \quad \dots(i)$$

Given, A(x, y) = A(1, 3)

$$x = X + 2 \text{ and } y = Y + 3 \quad \dots[\text{From (i)}]$$

$$\therefore 1 = X + 2 \text{ and } 3 = Y + 3$$

$\therefore X = -1$  and  $Y = 0$   
 $\therefore$  the new co-ordinates of point A are  $(-1, 0)$

### Exercise 5.1 | Q 7.2 | Page 67

If the origin is shifted to the point  $O'(2, 3)$ , the axes remaining parallel to the original axes, find the new co-ordinates of the points  $B(2, 5)$

#### **SOLUTION**

Origin is shifted to  $(2, 3) = (h, k)$   
Let the new co-ordinates be  $(X, Y)$ .  
 $\therefore x = X + h$  and  $y = Y + k$   
 $\therefore x = X + 2$  and  $y = Y + 3 \quad \dots(i)$   
Given,  $B(x, y) = B(2, 5)$   
 $x = X + 2$  and  $y = Y + 3 \quad \dots[\text{From (i)}]$   
 $\therefore 2 = X + 2$  and  $5 = Y + 3$   
 $\therefore X = 0$  and  $Y = 2$   
 $\therefore$  the new co-ordinates of point B are  $(0, 2)$ .

### Exercise 5.1 | Q 8.1 | Page 67

If the origin is shifted to the point  $O'(1, 3)$ , the axes remaining parallel to the original axes, find the old co-ordinates of the points  $C(5, 4)$

#### **SOLUTION**

Origin is shifted to  $(1, 3) = (h, k)$   
Let the new co-ordinates be  $(X, Y)$ .  
 $x = X + h$  and  $y = Y + k$   
 $\therefore x = X + 1$  and  $y = Y + 3 \quad \dots(i)$   
Given,  $C(X, Y) = C(5, 4)$   
 $x = X + 1$  and  $y = Y + 3 \quad \dots[\text{From (i)}]$   
 $\therefore x = 5 + 1 = 6$  and  $y = 4 + 3 = 7$   
 $\therefore$  the old co-ordinates of point C are  $(6, 7)$ .

### Exercise 5.1 | Q 8.2 | Page 67

If the origin is shifted to the point  $O'(1, 3)$ , the axes remaining parallel to the original axes, find the old co-ordinates of the points  $D(3, 3)$

#### **SOLUTION**

Origin is shifted to  $(1, 3) = (h, k)$   
Let the new co-ordinates be  $(X, Y)$ .  
 $x = X + h$  and  $y = Y + k$   
 $\therefore x = X + 1$  and  $y = Y + 3 \quad \dots(i)$   
Given,  $D(X, Y) = D(3, 3)$   
 $x = X + 1$  and  $y = Y + 3 \quad \dots[\text{From (i)}]$

$\therefore x = 3 + 1 = 4$  and  $y = 3 + 3 = 6$   
 $\therefore$  the old co-ordinates of point D are (4, 6).

### Exercise 5.1 | Q 9 | Page 67

If the co-ordinates (5, 14) change to (8, 3) by shift of origin, find the co-ordinates of the point, where the origin is shifted.

#### SOLUTION

Let the origin be shifted to (h, k).

Given,  $(x, y) = (5, 14)$ ,  $(X, Y) = (8, 3)$

Since,  $x = X + h$  and  $y = Y + k$

$\therefore 5 = 8 + h$  and  $14 = 3 + k$

$\therefore h = -3$  and  $k = 11$

$\therefore$  the co-ordinates of the point, where the origin is shifted are  $(-3, 11)$ .

### Exercise 5.1 | Q 10.1 | Page 67

Obtain the new equations of the following loci if the origin is shifted to the point  $O'(2, 2)$ , the direction of axes remaining the same:  $3x - y + 2 = 0$

#### SOLUTION

Given,  $(h, k) = (2, 2)$

Let  $(X, Y)$  be the new co-ordinates of the point  $(x, y)$ .

$\therefore x = X + h$  and  $y = Y + k$

$\therefore x = X + 2$  and  $y = Y + 2$

Substituting the values of  $x$  and  $y$  in the equation  $3x - y + 2 = 0$ , we get

$3(X + 2) - (Y + 2) + 2 = 0$

$\therefore 3X + 6 - Y - 2 + 2 = 0$

$\therefore 3X - Y + 6 = 0$ , which is the new equation of locus.

### Exercise 5.1 | Q 10.2 | Page 67

Obtain the new equations of the following loci if the origin is shifted to the point  $O'(2, 2)$ , the direction of axes remaining the same:  $x^2 + y^2 - 3x = 7$

#### SOLUTION

Given,  $(h, k) = (2, 2)$

Let  $(X, Y)$  be the new co-ordinates of the point  $(x, y)$ .

$\therefore x = X + h$  and  $y = Y + k$

$\therefore x = X + 2$  and  $y = Y + 2$

Substituting the values of  $x$  and  $y$  in the equation  $x^2 + y^2 - 3x = 7$ , we get

$(X + 2)^2 + (Y + 2)^2 - 3(X + 2) = 7$

$\therefore X^2 + 4X + 4 + Y^2 + 4Y + 4 - 3X - 6 = 7$

$\therefore X^2 + Y^2 + X + 4Y - 5 = 0$ , which is the new equation of locus.

### Exercise 5.1 | Q 10.3 | Page 67

Obtain the new equations of the following loci if the origin is shifted to the point  $O'(2, 2)$ , the direction of axes remaining the same:  $xy - 2x - 2y + 4 = 0$

#### **SOLUTION**

Given,  $(h, k) = (2, 2)$

Let  $(X, Y)$  be the new co-ordinates of the point  $(x, y)$ .

$$\therefore x = X + h \text{ and } y = Y + k$$

$$\therefore x = X + 2 \text{ and } y = Y + 2$$

Substituting the values of  $x$  and  $y$  in the equation  $xy - 2x - 2y + 4 = 0$ , we get

$$(X + 2)(Y + 2) - 2(X + 2) - 2(Y + 2) + 4 = 0$$

$$\therefore XY + 2X + 2Y + 4 - 2X - 4 - 2Y - 4 + 4 = 0$$

$$\therefore XY = 0, \text{ which is the new equation of locus.}$$

### **EXERCISE 5.2 [PAGES 69 - 70]**

### Exercise 5.2 | Q 1.1 | Page 69

Find the slope of the following lines which pass through the point:  $(2, -1)$ ,  $(4, 3)$

#### **SOLUTION**

Let  $A = (x_1, y_1) = (2, -1)$  and  $B = (x_2, y_2) = (4, 3)$ .

$$\text{Slope of line AB} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{3 - (-1)}{4 - 2}$$

$$= \frac{4}{2}$$

$$= 2.$$

### Exercise 5.2 | Q 1.2 | Page 69

Find the slope of the following lines which pass through the point:  $(-2, 3)$ ,  $(5, 7)$

#### **SOLUTION**

Let  $C = (x_1, y_1) = (-2, 3)$  and  $D = (x_2, y_2) = (5, 7)$ .

$$\begin{aligned}
 \text{Slope of line CD} &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{7 - 3}{5 - (-2)} \\
 &= \frac{4}{7}.
 \end{aligned}$$

### Exercise 5.2 | Q 1.3 | Page 69

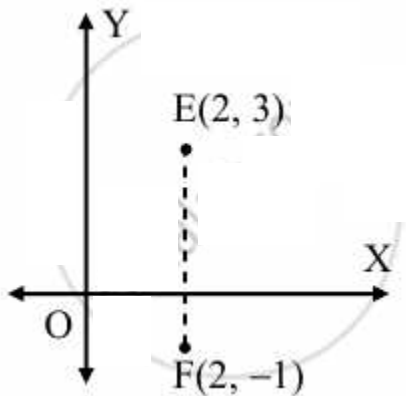
Find the slope of the following lines which pass through the point: (2, 3), (2, -1)

#### SOLUTION

Let E = (2, 3) = (x<sub>1</sub>, y<sub>1</sub>) and F = (2, -1) = (x<sub>2</sub>, y<sub>2</sub>)

Since x<sub>1</sub> = x<sub>2</sub> = 2

∴ The slope of EF is not defined. ...[EF || y-axis]



### Exercise 5.2 | Q 1.4 | Page 69

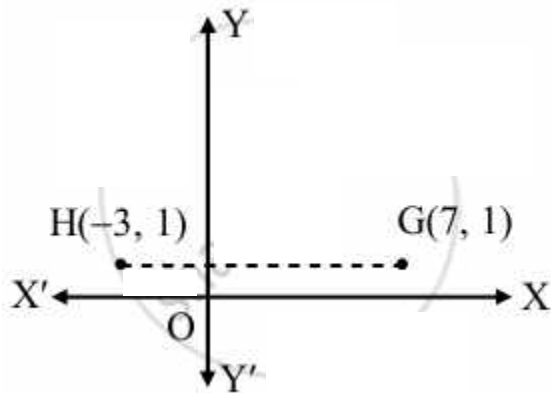
Find the slope of the following lines which pass through the point: (7, 1), (-3, 1)

#### SOLUTION

Let, G = (7, 1) = (x<sub>1</sub>, y<sub>1</sub>) and H = (-3, 1) = (x<sub>2</sub>, y<sub>2</sub>) say.

Since y<sub>1</sub> = y<sub>2</sub>

∴ The slope of GH = 0. ...[GH || x-axis]



### Exercise 5.2 | Q 2 | Page 69

If the X and Y-intercepts of line L are 2 and 3 respectively, then find the slope of line L.

#### SOLUTION

Given,

x-intercept of line L is 2 and

y-intercept of line L is 3

∴ the line L intersects X-axis at (2, 0) and Y-axis at (0, 3).

i.e. the line L passes through (2, 0) = (x<sub>1</sub>, y<sub>1</sub>) and (0, 3) = (x<sub>2</sub>, y<sub>2</sub>) say.

$$\text{Slope of line L} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{3 - 0}{0 - 2}$$

$$= \frac{-3}{2}.$$

### Exercise 5.2 | Q 3 | Page 69

Find the slope of the line whose inclination is 30°.

#### SOLUTION

Given, inclination (θ) = 30°

$$\text{Slope of the line} = \tan \theta = \tan 30^\circ = \frac{1}{\sqrt{3}}.$$

### Exercise 5.2 | Q 4 | Page 69

Find the slope of the line whose inclination is 45°.

**SOLUTION**

Given, inclination  $(\theta) = 45^\circ$

Slope of the line  $= \tan \theta = \tan 45^\circ = 1$ .

**Exercise 5.2 | Q 5 | Page 69**

A line makes intercepts 3 and 3 on the co-ordinate axes. Find the slope of the line.

**SOLUTION**

Given,

x-intercept of line is 3 and

y-intercept of line is 3

$\therefore$  The line intersects X-axis at (3, 0) and Y-axis at (0, 3).

i.e. the line passes through (3, 0) =  $(x_1, y_1)$  and (0, 3) =  $(x_2, y_2)$  say.

Given, inclination  $(\theta) = 30^\circ$

$$\text{Slope of the line} = \tan \theta = \tan 30^\circ = \frac{1}{\sqrt{3}}.$$

**Exercise 5.2 | Q 6 | Page 69**

Without using Pythagoras theorem, show that points A (4, 4), B (3, 5) and C (− 1, − 1) are the vertices of a right-angled triangle.

**SOLUTION**

Given, A(4, 4) =  $(x_1, y_1)$ , B(3, 5) =  $(x_2, y_2)$ , C (− 1, − 1) =  $(x_3, y_3)$

$$\text{Slope of AB} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{5 - 4}{3 - 4} = -1$$

$$\text{Slope of BC} = \frac{y_3 - y_2}{x_3 - x_2} = \frac{-1 - 5}{-1 - 3} = \frac{-6}{-4} = \frac{3}{2}$$

$$\text{Slope of AC} = \frac{y_3 - y_1}{x_3 - x_1} = \frac{-1 - 4}{-1 - 4} = \frac{-5}{-5} = 1$$

$$\text{Slope of AB} \times \text{slope of AC} = -1 \times 1 = -1$$

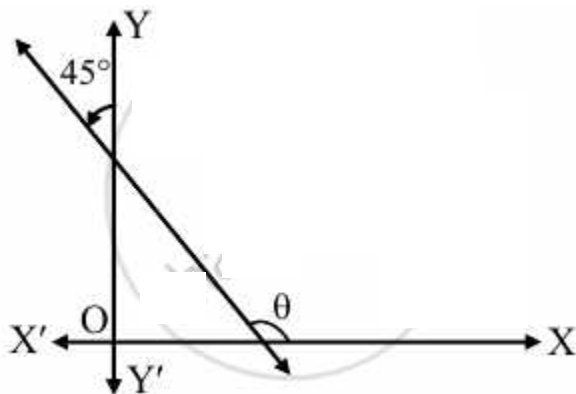
$\therefore$  side AB  $\perp$  side AC

$\therefore$   $\triangle ABC$  is a right angled triangle, right angled at A.

$\therefore$  The given points are the vertices of a right angled triangle.

**Exercise 5.2 | Q 7 | Page 69**

Find the slope of the line which makes angle of  $45^\circ$  with the positive direction of the Y-axis measured clockwise.

**SOLUTION**

Since, the line makes an angle of  $45^\circ$  with positive direction of Y-axis in anticlockwise direction.

$$\therefore \text{Inclination of the line } (\theta) = (90^\circ + 45^\circ)$$

$$\therefore \text{Slope of the line} = \tan(90^\circ + 45^\circ)$$

$$= -\cot 45^\circ \quad \dots [\tan(90^\circ + \theta) = -\cot \theta]$$

$$= -1.$$

**Exercise 5.2 | Q 8 | Page 70**

Find the value of k for which the points P(k, -1), Q(2, 1) and R(4, 5) are collinear.

**SOLUTION**

Given, points P(k, -1), Q(2, 1) and R(4, 5) are collinear.

$$\therefore \text{Slope of PQ} = \text{Slope of QR}$$

$$\therefore \frac{1 - (-1)}{2 - k} = \frac{5 - 1}{4 - 2}$$

$$\therefore \frac{2}{2 - k} = \frac{4}{2}$$

$$\therefore 1 = 2 - k$$

$$\therefore k = 2 - 1 = 1.$$

**EXERCISE 5.3 [PAGE 73]****Exercise 5.3 | Q 1.1 | Page 73**

Write the equation of the line: parallel to the X-axis and at a distance of 5 units from it and above it.

**SOLUTION**

Equation of a line parallel to X-axis is  $y = k$ .

Since, the line is at a distance of 5 units above X-axis.

$$\therefore k = 5$$

$\therefore$  the equation of the required line is  $y = 5$ .

**Exercise 5.3 | Q 1.2 | Page 73**

Write the equation of the line: parallel to the Y-axis and at a distance of 5 units from it and to the left of it.

**SOLUTION**

Equation of a line parallel to Y-axis is  $x = h$ .

Since, the line is at a distance of 5 units to the left of Y-axis.

$$\therefore h = -5$$

$\therefore$  the equation of the required line is  $x = -5$ .

**Exercise 5.3 | Q 1.3 | Page 73**

Write the equation of the line: parallel to the X-axis and at a distance of 4 units from the point  $(-2, 3)$ .

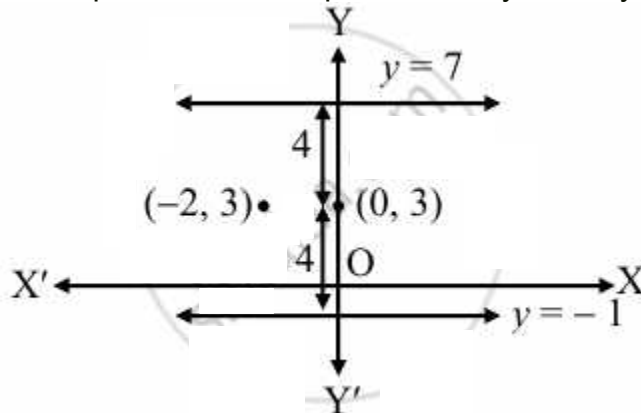
**SOLUTION**

Equation of a line parallel to the X-axis is of the form  $y = k$  ( $k > 0$  or  $k < 0$ ).

Since, the line is at a distance of 4 units from the point  $(-2, 3)$ .

$$\therefore k = 3 + 4 = 7 \text{ or } k = 3 - 4 = -1$$

$\therefore$  the equation of the required line is  $y = 7$  or  $y = -1$ .

**Exercise 5.3 | Q 2.1 | Page 73**

Obtain the equation of the line: parallel to the X-axis and making an intercept of 3 units on the Y-axis.

**SOLUTION**

Equation of a line parallel to X-axis with y-intercept 'k' is  $y = k$ .

Here, y-intercept = 3

$\therefore$  the equation of the required line is  $y = 3$ .

**Exercise 5.3 | Q 2.2 | Page 73**

Obtain the equation of the line: parallel to the Y-axis and making an intercept of 4 units on the X-axis.

**SOLUTION**

Equation of a line parallel to Y-axis with x-intercept 'h' is  $x = h$ .

Here, x-intercept = 4

$\therefore$  the equation of the required line is  $x = 4$ .

**Exercise 5.3 | Q 3.1 | Page 73**

Obtain the equation of the line containing the point: A(2, -3) and parallel to the Y-axis.

**SOLUTION**

Equation of a line parallel to Y-axis is of the form  $x = h$ .

Since, the line passes through A(2, -3).

$\therefore h = 2$

$\therefore$  the equation of the required line is  $x = 2$ .

**Exercise 5.3 | Q 3.2 | Page 73**

Obtain the equation of the line containing the point: B(4, -3) and parallel to the X-axis.

**SOLUTION**

Equation of a line parallel to X-axis is of the form  $y = k$ .

Since, the line passes through B(4, -3)

$\therefore k = -3$

$\therefore$  the equation of the required line is  $y = -3$ .

**Exercise 5.3 | Q 4 | Page 73**

Find the equation of the line passing through the points A(2, 0) and B(3, 4).

**SOLUTION**

The required line passes through the points A(2, 0) =  $(x_1, y_1)$  and B(3, 4) =  $(x_2, y_2)$  say.

Equation of the line in two point form is

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$\therefore$  the equation of the required line is

$$\frac{y - 0}{4 - 0} = \frac{x - 2}{3 - 2}$$

$$\therefore \frac{y}{4} = \frac{x - 2}{1}$$

$$\therefore y = 4(x - 2)$$

$$\therefore y = 4x - 8$$

$$\therefore 4x - y - 8 = 0.$$

### Exercise 5.3 | Q 5 | Page 73

Line  $y = mx + c$  passes through the points A(2, 1) and B(3, 2). Determine  $m$  and  $c$ .

#### **SOLUTION**

Given, A(2, 1) and B(3, 2).

Equation of a line in two point form is

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$\therefore$  the equation of the passing through A and B line is

$$\frac{y - 1}{2 - 1} = \frac{x - 2}{3 - 2}$$

$$\therefore \frac{y - 1}{1} = \frac{x - 2}{1}$$

$$\therefore y - 1 = x - 2$$

$$\therefore y = x - 1$$

Comparing this equation with  $y = mx + c$ , we get

$$m = 1 \text{ and } c = -1$$

#### **Alternative method:**

Points A(2, 1) and B(3, 2) lie on the line  $y = mx + c$ .

$\therefore$  They must satisfy the equation.

$$\therefore 2m + c = 1 \quad \dots(i)$$

$$\text{and } 3m + c = 2 \quad \dots(ii)$$

equation (ii) equation (i) gives  $m = 1$

Substituting  $m = 1$  in (i), we get

$$2(1) + c = 1$$

$$\therefore c = 1 - 2 = -1.$$

### Exercise 5.3 | Q 6.1 | Page 73

The vertices of a triangle are A(3, 4), B(2, 0) and C(-1, 6). Find the equations of side BC

#### SOLUTION

Vertices of  $\triangle ABC$  are A(3, 4), B(2, 0) and C(-1, 6).

Equation of a line in two point form is

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

$\therefore$  the equation of the side BC is

$$\frac{y - 0}{6 - 0} = \frac{x - 2}{-1 - 2} \quad \dots \left[ \begin{array}{l} B = (x_1, y_1) = (2, 0) \\ C = (x_2, y_2) = (-1, 6) \end{array} \right]$$

$$\therefore \frac{y}{6} = \frac{x - 2}{-3}$$

$$y = -2(x - 2)$$

$$\therefore 2x + y - 4 = 0.$$

### Exercise 5.3 | Q 6.2 | Page 73

The vertices of a triangle are A(3, 4), B(2, 0) and C(-1, 6). Find the equations of the median AD.

#### SOLUTION

Vertices of  $\triangle ABC$  are A(3, 4), B(2, 0) and C(-1, 6).

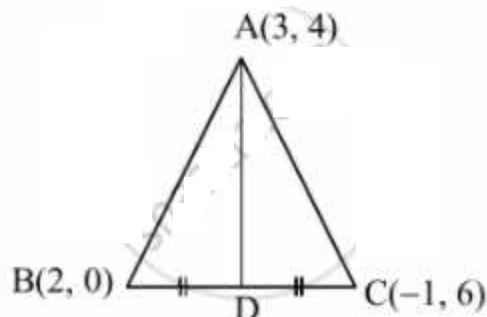
Let D be the midpoint of side BC.

Then, AD is the median through A.

$$\therefore D = \left( \frac{2-1}{2}, \frac{0+6}{2} \right) = \left( \frac{1}{2}, 3 \right)$$

The median AD passes through the points

A(3, 4) and D $\left(\frac{1}{2}, 3\right)$ .



$\therefore$  the equation of the median AD is

$$\frac{y-4}{3-4} = \frac{x-3}{\frac{1}{2}-3}$$

$$\therefore \frac{y-4}{-1} = \frac{x-3}{-\frac{5}{2}}$$

$$\therefore \frac{5}{2}(y-4) = x-3$$

$$\therefore 5y - 20 = 2x - 6$$

$$\therefore 2x - 5y + 14 = 0.$$

### Exercise 5.3 | Q 6.3 | Page 73

The vertices of a triangle are A(3, 4), B(2, 0) and C(-1, 6). Find the equations of the midpoints of sides AB and BC.

#### **SOLUTION**

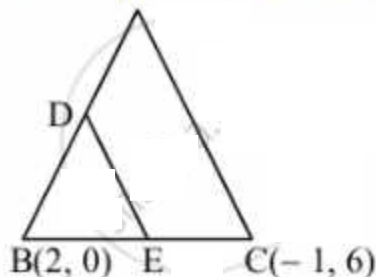
Vertices of  $\triangle ABC$  are A(3, 4), B(2, 0) and C(-1, 6).

Let D and E be the midpoints of side AB and side BC respectively.

$$\therefore D = \left( \frac{3+2}{2}, \frac{4+0}{2} \right) = \left( \frac{5}{2}, 2 \right) \text{ and}$$

$$E = \left( \frac{2-1}{2}, \frac{0+6}{2} \right) = \left( \frac{1}{2}, 3 \right)$$

∴ the equation of the line DE is  $A(3, 4)$



$$\frac{y-2}{3-2} = \frac{x-\frac{5}{2}}{\frac{1}{2}-\frac{5}{2}}$$

$$\therefore \frac{y-2}{1} = \frac{2x-5}{-4}$$

$$\therefore -4(y-2) = 2x-5$$

$$\therefore -4y+8 = 2x-5$$

$$\therefore 2x+4y-13 = 0.$$

Exercise 5.3 | Q 7.1 | Page 73

Find the x and y-intercepts of the following line:  $\frac{x}{y} + \frac{y}{2} = 1$

**SOLUTION**

Given equation of the line is  $\frac{x}{y} + \frac{y}{2} = 1$

This is of the form  $\frac{x}{a} + \frac{y}{b} = 1$ ,

where x-intercept = a, y-intercept = b

∴ x-intercept = 3, y-intercept = 2.

Exercise 5.3 | Q 7.2 | Page 73

Find the x and y-intercepts of the following line:  $\frac{3x}{2} + \frac{2y}{3} = 1$

**SOLUTION**

Given equation of the line is  $\frac{3x}{2} + \frac{2y}{3} = 1$

$$\therefore \frac{x}{\left(\frac{2}{3}\right)} + \frac{y}{\left(\frac{3}{2}\right)} = 1$$

This is of the form  $\frac{x}{a} + \frac{y}{b} = 1$ ,

where x-intercept = a, y-intercept = b

$$\therefore \text{x-intercept} = \frac{2}{3} \text{ and y-intercept} = \frac{3}{2}.$$

**Exercise 5.3 | Q 7.3 | Page 73**

Find the x and y-intercepts of the following line:  $2x - 3y + 12 = 0$

**SOLUTION**

Given equation of the line is  $2x - 3y + 12 = 0$

$$\therefore 2x - 3y = -12$$

$$\therefore \frac{2x}{(-12)} - \frac{3y}{(-12)} = 1$$

$$\therefore \frac{x}{-6} + \frac{y}{4} = 1$$

This is of the form  $\frac{x}{a} + \frac{y}{b} = 1$ ,

where x-intercept = a, y-intercept = b

$$\therefore \text{x-intercept} = -6 \text{ and y-intercept} = 4.$$

**Exercise 5.3 | Q 8 | Page 73**

Find the equations of a line containing the point A(3, 4) and making equal intercepts on the co-ordinate axes.

**SOLUTION**

Let the equation of the line be

$$\frac{x}{a} + \frac{y}{b} = 1$$

Since, the required line make equal intercepts on the co-ordinate axes.

$$\therefore a = b$$

$$\therefore \text{(i) reduces to } x + y = a \quad \dots\text{(ii)}$$

Since the line passes through A(3, 4).

$$\therefore 3 + 4 = a$$

$$\text{i.e. } a = 7$$

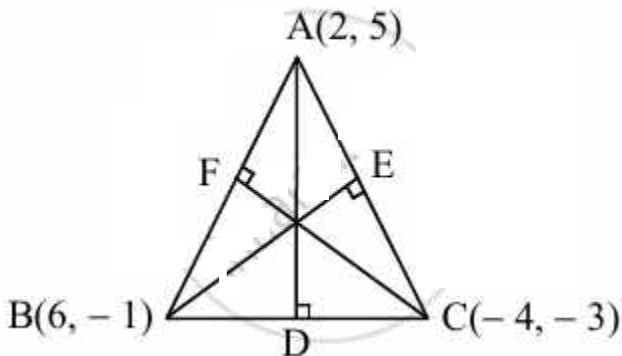
Substituting  $a = 7$  in (ii) to get

$$x + y = 7.$$

### Exercise 5.3 | Q 9 | Page 73

Find the equations of the altitudes of the triangle whose vertices are A(2, 5), B(6, -1) and C(-4, -3).

#### SOLUTION



A(2, 5), B(6, -1), C(-4, -3) are the vertices of  $\triangle ABC$ .

Let AD, BE and CF be the altitudes through the vertices A, B and C respectively of  $\triangle ABC$ .

$$\text{Slope of BC} = \frac{-3 - (-1)}{4 - 6} = \frac{-2}{-10} = \frac{1}{5}$$

$$\therefore \text{slope of AD} = -5 \quad \dots[\because AD \perp BC]$$

Since, altitude AD passes through (2, 5) and has slope -5.

$\therefore$  the equation of the altitude AD is

$$y - 5 = -5(x - 2)$$

$$\therefore y - 5 = -5x + 10$$

$$\therefore 5x + y - 15 = 0$$

$$\text{Now, slope of AC} = \frac{-3 - 5}{-4 - 2} = \frac{-8}{-6} = \frac{4}{3}$$

$$\therefore \text{Slope of BE} = \frac{-3}{4} \quad \dots[\because \text{BE} \perp \text{AC}]$$

Since, altitude BE passes through  $(6, -1)$  and has slope  $\frac{-3}{4}$ .

$\therefore$  the equation of the altitude BE is

$$y - (-1) = \frac{-3}{4}(x - 6)$$

$$\therefore 4(y + 1) = -3(x - 6)$$

$$\therefore 3x + 4y - 14 = 0$$

$$\text{Also, slope of AB} = \frac{-1 - 5}{6 - 2} = \frac{-6}{4} = \frac{-3}{2}$$

$$\therefore \text{Slope of BE} = \frac{2}{3} \quad \dots[\because \text{CF} \perp \text{AB}]$$

Since, altitude CF passes through  $(-4, -3)$  and has slope  $\frac{2}{3}$ .

$\therefore$  the equation of the altitude CF is

$$y - (-3) = \frac{2}{3}[x - (-4)]$$

$$\therefore 3(y + 3) = 2(x + 4)$$

$$\therefore 2x - 3y - 1 = 0.$$

#### EXERCISE 5.4 [PAGE 78]

##### Exercise 5.4 | Q 1.1 | Page 78

Find the slope, x-intercept, y-intercept of the following line :  $2x + 3y - 6 = 0$

**SOLUTION**

Given equation of the line is  $2x + 3y - 6 = 0$

Comparing this equation with  $ax + by + c = 0$ , we get

$$a = 2, b = 3, c = -6$$

$$\therefore \text{Slope of the line} = \frac{-a}{b} = \frac{-2}{3}$$

$$\text{x-intercept} = \frac{-c}{a} = \frac{-(-6)}{2} = 3$$

$$\text{y-intercept} = \frac{-c}{b} = \frac{-(-6)}{3} = 2$$

**Exercise 5.4 | Q 1.2 | Page 78**

Find the slope, x-intercept, y-intercept of the following line :  $x + 2y = 0$

**SOLUTION**

Given equation of the line is  $x + 2y = 0$

Comparing this equation with  $ax + by + c = 0$ ,

we get

$$a = 1, b = 2, c = 0$$

$$\therefore \text{Slope of the line} = \frac{-a}{b} = \frac{-1}{2}$$

$$\text{x-intercept} = \frac{-c}{a} = \frac{0}{1} = 0$$

$$\text{y-intercept} = \frac{-c}{b} = \frac{0}{2} = 0$$

**Exercise 5.4 | Q 2.1 | Page 78**

Write the following equation in  $ax + by + c = 0$  form:  $y = 2x - 4$

**SOLUTION**

$$y = 2x - 4$$

$\therefore 2x - y - 4 = 0$  is the equation in  $ax + by + c = 0$  form.

**Exercise 5.4 | Q 2.2 | Page 78**

Write the following equation in  $ax + by + c = 0$  form:  $y = 4$

**SOLUTION**

$$y = 4$$

$\therefore 0x + 1y - 4 = 0$  is the equation in  $ax + by + c = 0$  form.

**Exercise 5.4 | Q 2.3 | Page 78**

Write the following equation in  $ax + by + c = 0$  form:  $\frac{x}{2} + \frac{y}{4} = 1$

**SOLUTION**

$$\frac{x}{2} + \frac{y}{4} = 1$$

$$\therefore \frac{2x + y}{4} = 1$$

$$\therefore 2x + y = 4$$

$\therefore 2x + y - 4 = 0$  is the equation in  $ax + by + c = 0$  form.

**Exercise 5.4 | Q 2.4 | Page 78**

Write the following equation in  $ax + by + c = 0$  form:  $\frac{x}{3} = \frac{y}{2}$

**SOLUTION**

$$\frac{x}{3} = \frac{y}{2}$$

$$\therefore 2x = 3y$$

$\therefore 2x - 3y + 0 = 0$  is the equation in  $ax + by + c = 0$  form.

**Exercise 5.4 | Q 3 | Page 78**

Show that the lines  $x - 2y - 7 = 0$  and  $2x - 4y + 5 = 0$  are parallel to each other.

**SOLUTION**

Let  $m_1$  be the slope of the line  $x - 2y - 7 = 0$ .

$$\therefore m_1 = \frac{-2}{-2} = \frac{1}{2}$$

Let  $m_2$  be the slope of the line  $2x - 4y + 5 = 0$ .

$$\therefore m_2 = \frac{-2}{-4} = \frac{1}{2}$$

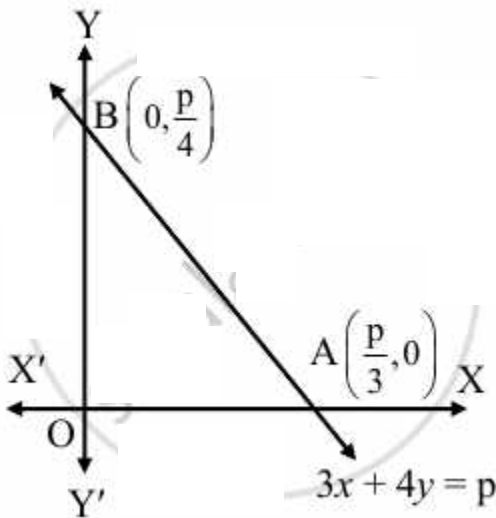
Since,  $m_1 = m_2$

$\therefore$  The given lines are parallel to each other.

#### Exercise 5.4 | Q 4 | Page 78

If the line  $3x + 4y = p$  makes a triangle of area 24 square units with the co-ordinate axes, then find the value of  $p$ .

#### SOLUTION



Let the line  $3x + 4y = p$  cuts the X and Y-axes at points A and B respectively.

$$3x + 4y = p$$

$$\therefore \frac{3x}{p} + \frac{4y}{p} = 1$$

$$\therefore \frac{x}{\frac{p}{3}} + \frac{y}{\frac{p}{4}} = 1$$

This equation is of the form

$$\frac{x}{a} + \frac{y}{b} = 1.$$

$$\text{with } a = \frac{p}{3} \text{ and } b = \frac{p}{4}$$

$$\therefore A = (a, 0) = \left(\frac{P}{3}, 0\right) \text{ and } B = (0, b) = \left(0, \frac{P}{4}\right)$$

$$\therefore OA = \frac{P}{4} \text{ and } OB = \frac{P}{4}$$

Given,  $A(\Delta.OAB) = 24$  sq. units

$$\therefore \frac{1}{2} \times OA \times OB = 24$$

$$\therefore \frac{1}{2} \times \frac{P}{3} \times \frac{P}{4} = 24$$

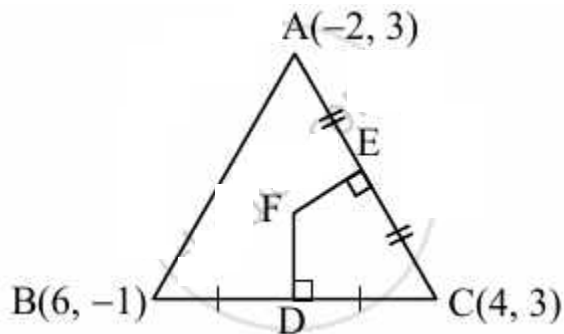
$$\therefore p^2 = 576$$

$$\therefore p = \pm 24.$$

#### Exercise 5.4 | Q 5 | Page 78

Find the co-ordinates of the circumcentre of the triangle whose vertices are  $A(-2, 3)$ ,  $B(6, -1)$ ,  $C(4, 3)$ .

#### SOLUTION



Here,  $A(-2, 3)$ ,  $B(6, -1)$ ,  $C(4, 3)$  are the vertices of  $\Delta ABC$ .  
Let  $F$  be the circumcentre of  $\Delta ABC$ .

Let  $FD$  and  $FE$  be the perpendicular bisectors of the sides  $BC$  and  $AC$  respectively.  
 $\therefore D$  and  $E$  are the midpoints of side  $BC$  and  $AC$  respectively.

$$\therefore D = \left(\frac{6+4}{2}, \frac{-1+3}{2}\right)$$

$$\therefore D = (5, 1) \text{ and } E = \left(\frac{-2+4}{2}, \frac{3+3}{2}\right)$$

$$\therefore E = (1, 3)$$

$$\text{Now, slope of BC} = \frac{-1 - 3}{6 - 4} = -2$$

$$\therefore \text{slope of FD} = \frac{1}{2} \quad \dots[\because \text{FD} \perp \text{BC}]$$

Since, FD passes through (5, 1) and has slope  $\frac{1}{2}$

$$\therefore \text{Equation of FD is } y - 1 = \frac{1}{2}(x - 5)$$

$$\therefore 2(y - 1) = x - 5$$

$$\therefore x - 2y - 3 = 0 \quad \dots(i)$$

Since, both the points A and C have same y co-ordinates i.e. 3

$\therefore$  the points A and C lie on the line  $y = 3$ .

Since, FE passes through E(1, 3).

$\therefore$  the equation of FE is  $x = 1$ . ... (ii)

To find co-ordinates of circumcentre, we have to solve equations (i) and (ii).

Substituting the value of x in (i), we get

$$1 - 2y - 3 = 0$$

$$\therefore y = -1$$

$\therefore$  Co-ordinates of circumcentre F  $\equiv (1, -1)$ .

#### Exercise 5.4 | Q 6 | Page 78

Find the equation of the line whose x-intercept is 3 and which is perpendicular to the line  $3x - y + 23 = 0$ .

#### **SOLUTION**

Slope of the line  $3x - y + 23 = 0$  is 3.

$\therefore$  slope of the required line which is perpendicular to  $3x - y + 23 = 0$  is  $-1/3$

Since, the x-intercept of the required line is 3.

$\therefore$  it passes through (3, 0).

∴ the equation of the required line is

$$y - 0 = \frac{-1}{3}(x - 3)$$

$$\therefore 3y = -x + 3$$

$$\therefore x + 3y = 3.$$

**Exercise 5.4 | Q 7 | Page 78**

Find the distance of the point A(- 2, 3) from the line  $12x - 5y - 13 = 0$ .

**SOLUTION**

Let p be the perpendicular distance of the point A(- 2, 3) from the line  $12x - 5y - 13 = 0$   
Here,  $a = 12$ ,  $b = -5$ ,  $c = -13$ ,  $x_1 = -2$ ,  $y_1 = 3$

$$\begin{aligned}\therefore p &= \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right| \\&= \left| \frac{12(-2) - 5(3) - 13}{\sqrt{12^2 + (-5)^2}} \right| \\&= \left| \frac{-24 - 15 - 13}{\sqrt{144 + 25}} \right| \\&= \left| \frac{-52}{13} \right| \\&= 4 \text{ units.}\end{aligned}$$

**Exercise 5.4 | Q 8 | Page 78**

Find the distance between parallel lines  $9x + 6y - 7 = 0$  and  $9x + 6y - 32 = 0$ .

**SOLUTION**

Equations of the given parallel lines are  $9x + 6y - 7 = 0$  and  $9x + 6y - 32 = 0$ .

Here,  $a = 9$ ,  $b = 6$ ,  $C_1 = -7$  and  $C_2 = -32$

∴ Distance between the parallel lines

$$\begin{aligned}
&= \left| \frac{C_1 - C_2}{\sqrt{a^2 + b^2}} \right| \\
&= \left| \frac{-7 - (-32)}{\sqrt{9^2 + 6^2}} \right| \\
&= \left| \frac{-7 + 32}{\sqrt{81 + 36}} \right| \\
&= \left| \frac{25}{\sqrt{117}} \right| \\
&= \frac{25}{\sqrt{117}} \text{ units.}
\end{aligned}$$

#### Exercise 5.4 | Q 9 | Page 78

Find the equation of the line passing through the point of intersection of lines  $x + y - 2 = 0$  and  $2x - 3y + 4 = 0$  and making intercept 3 on the X-axis.

#### **SOLUTION**

Given equations of lines are

$$\begin{aligned}
x + y - 2 &= 0 && \dots(i) \\
\text{and } 2x - 3y + 4 &= 0 && \dots(ii)
\end{aligned}$$

Multiplying equation (i) by 3, we get  
 $3x + 3y - 6 = 0$  ...(iii)

Adding equation (ii) and (iii), we get  
 $5x - 2 = 0$

$$\therefore x = \frac{2}{5}$$

Substituting  $x = \frac{2}{5}$  in equation (i), we get

$$\frac{2}{5} + y - 2 = 0$$

$$\therefore y = 2 - \frac{2}{5} = \frac{8}{5}$$

$\therefore$  The required line passes through point  $\left(\frac{2}{5}, \frac{8}{5}\right)$ .

Also, the line makes intercept of 3 on X-axis

$\therefore$  it also passes through point (3, 0).

$\therefore$  required equation of line passing through points  $\left(\frac{2}{5}, \frac{8}{5}\right)$  and (3, 0) is

$$\frac{y - \frac{8}{5}}{0 - \frac{8}{5}} = \frac{x - \frac{2}{5}}{3 - \frac{2}{5}}$$

$$\therefore \frac{\frac{5y-8}{5}}{-\frac{8}{5}} = \frac{\frac{5x-2}{5}}{\frac{13}{5}}$$

$$\therefore \frac{5y-8}{-8} = \frac{5x-2}{13}$$

$$\therefore 13(5y-8) = -8(5x-2)$$

$$\therefore 65y - 104 = -40x + 16$$

$$\therefore 40x + 65y - 120 = 0$$

$$\therefore 8x + 13y - 24 = 0 \text{ which is the equation of the required line.}$$

#### Exercise 5.4 | Q 10.1 | Page 78

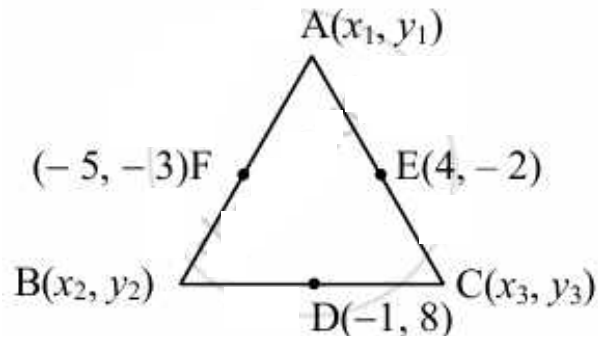
D(-1, 8), E(4, -2), F(-5, -3) are midpoints of sides BC, CA and AB of  $\Delta ABC$

Find: equations of sides of  $\Delta ABC$

#### **SOLUTION**

Let A( $x_1$ ,  $y_1$ ), B( $x_2$ ,  $y_2$ ) and C( $x_3$ ,  $y_3$ ) be the vertices of  $\Delta ABC$ .

Given, points D, E and F are midpoints of sides BC, CA and AB respectively of  $\Delta ABC$ .



$$D = \left( \frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2} \right)$$

$$\therefore (-1, 8) = \left( \frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2} \right)$$

$$\therefore x_2 + x_3 = -2 \quad \dots(i)$$

$$\text{and } y_2 + y_3 = 16 \quad \dots(ii)$$

$$\text{Also, } E = \left( \frac{x_1 + x_3}{2}, \frac{y_1 + y_3}{2} \right)$$

$$\therefore (4, -2) = \left( \frac{x_1 + x_3}{2}, \frac{y_1 + y_3}{2} \right)$$

$$\therefore x_1 + x_3 = 8 \quad \dots(iii)$$

$$\text{and } y_1 + y_3 = -4 \quad \dots(iv)$$

$$\text{Similarly, } F = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\therefore (-5, -3) = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\therefore x_1 + x_2 = -10 \quad \dots(v)$$

$$\text{and } y_1 + y_2 = -6 \quad \dots(vi)$$

**For x-coordinates:**

Adding (i), (iii) and (v), we get

$$2x_1 + 2x_2 + 2x_3 = -4$$

$$\therefore x_1 + x_2 + x_3 = -2$$

Solving (i) and (vii), we get

$$x_1 = 0$$

Solving (iii) and (vii), we get

$$x_2 = -10$$

Solving (v) and (vii), we get

$$x_3 = 8$$

**For y-coordinates:**

Adding (ii), (iv) and (vi), we get

$$2y_1 + 2y_2 + 2y_3 = 6$$

$$\therefore y_1 + y_2 + y_3 = 3 \quad \dots(\text{viii})$$

Solving (ii) and (viii), we get

$$y_1 = -13$$

Solving (iv) and (viii), we get

$$y_2 = 7$$

Solving (vi) and (viii), we get

$$y_3 = 9$$

$\therefore$  Vertices of  $\Delta ABC$  are A.(0, -13), B(-10, 7), C(8, 9)

**a.** Equation of side AB is

$$\frac{y + 13}{7 + 13} = \frac{x - 0}{-10 - 0}$$

$$\therefore \frac{y + 13}{20} = \frac{x}{-10}$$

$$\therefore \frac{y + 13}{2} = -x$$

$$\therefore 2x + y + 13 = 0$$

**b.** Equation of side BC is

$$\frac{y - 7}{9 - 7} = \frac{x + 10}{8 + 10}$$

$$\therefore \frac{y - 7}{2} = \frac{x + 10}{9}$$

$$\therefore y - 7 = \frac{x + 10}{9}$$

$$\therefore x - 9y + 73 = 0$$

**c.** Equation of side AC is

$$\frac{y + 13}{9 + 13} = \frac{x - 0}{8 - 0}$$

$$\therefore \frac{y + 13}{22} = \frac{x}{8}$$

$$\therefore 8(y + 13) = 22x$$

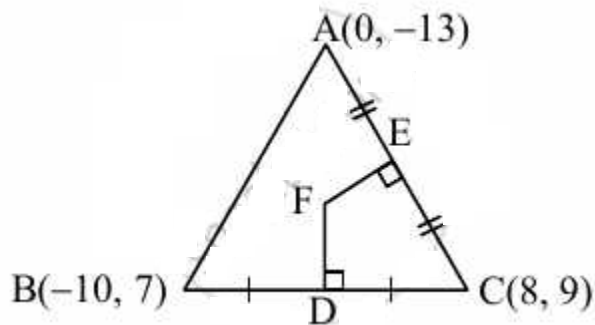
$$\therefore 4(y + 13) = 11x$$

$$\therefore 11x - 4y - 52 = 0.$$

### Exercise 5.4 | Q 10.2 | Page 78

D(-1, 8), E(4, -2), F(-5, -3) are midpoints of sides BC, CA and AB of  $\Delta ABC$ . Find: coordinates of the circumcentre of  $\Delta ABC$ .

#### SOLUTION



Here, A(0, -13), B(-10, 7), C(8, 9) are the vertices of  $\Delta ABC$ .  
Let F be the circumcentre of  $\Delta ABC$ .

Let FD and FE be perpendicular bisectors of the sides BC and AC respectively.  
 $\therefore$  D and E are the midpoints of side BC and AC.

$$\therefore D = \left( \frac{-10 + 8}{2}, \frac{7 + 9}{2} \right)$$

$$\therefore D = (-1, 8) \text{ and } E = \left( \frac{0 + 8}{2}, \frac{-13 + 9}{2} \right)$$

$$\therefore E = (4, -2)$$

$$\text{Now, slope of BC} = \frac{7 - 9}{-10 - 8} = \frac{1}{9}$$

$$\therefore \text{slope of FD} = -9 \quad \dots [\because \text{FD} \perp \text{BC}]$$

Since, FD passes through (-1, 8) and has slope -9

$\therefore$  Equation of FD is

$$y - 8 = -9(x + 1)$$

$$\therefore y - 8 = -9x - 9$$

$$\therefore y = -9x - 1 \quad \dots(i)$$

$$\text{Also, slope of AC} = \frac{-13 - 9}{0 - 8} = \frac{11}{4}$$

$$\therefore \text{Slope of FE} = \frac{-4}{11} \quad \dots[\because \text{FE} \perp \text{AC}]$$

Since, FE passes through  $(4, -2)$  and has slope  $\frac{-4}{11}$

$\therefore$  Equation of FE is

$$y + 2 = \frac{-4}{11}(x - 4)$$

$$\therefore 11(y + 2) = -4(x - 4)$$

$$\therefore 11y + 22 = -4x + 16$$

$$\therefore 4x + 11y = -6 \quad \dots(ii)$$

To find co-ordinates of circumcentre,

we have to solve equations (i) and (ii).

Substituting the value of  $y$  in (ii), we get

$$4x + 11(-9x - 1) = -6$$

$$\therefore 4x - 99x - 11 = -6$$

$$\therefore -95x = 5$$

$$\therefore x = \frac{-1}{19}$$

Substituting the value of  $x$  in (i), we get

$$y = -9\left(-\frac{1}{19}\right) - 1 = \frac{-10}{19}$$

$$\therefore \text{Co-ordinates of circumcentre F} \equiv \left(\frac{-1}{19}, \frac{-10}{19}\right).$$

**MISCELLANEOUS EXERCISE 5 [PAGES 79 - 80]****Miscellaneous Exercise 5 | Q 1.1 | Page 79**

Find the slope of the line passing through the following point: (1, 2), (3, -5)

**SOLUTION**

Let A = (1, 2) = (x<sub>1</sub>, y<sub>1</sub>) and B = (3, -5) = (x<sub>2</sub>, y<sub>2</sub>) say.

$$\begin{aligned}\text{Slope of line AB} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{-5 - 2}{3 - 1} \\ &= \frac{-7}{2}.\end{aligned}$$

**Miscellaneous Exercise 5 | Q 1.2 | Page 79**

Find the slope of the line passing through the following point: (1, 3), (5, 2)

**SOLUTION**

Let C = (1, 3) = (x<sub>1</sub>, y<sub>1</sub>) and D = (5, 2) = (x<sub>2</sub>, y<sub>2</sub>) say.

$$\begin{aligned}\text{Slope of line CD} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{2 - 3}{5 - 1} \\ &= \frac{-1}{4}.\end{aligned}$$

**Miscellaneous Exercise 5 | Q 1.3 | Page 79**

Find the slope of the line passing through the following point: (-1, 3), (3, -1)

**SOLUTION**

Let E = (-1, 3) = (x<sub>1</sub>, y<sub>1</sub>) and F = (3, -1) = (x<sub>2</sub>, y<sub>2</sub>) say.

$$\begin{aligned}\text{Slope of line EF} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{-1 - 3}{3 - (-1)}\end{aligned}$$

$$= \frac{-4}{4}$$

$$= -1.$$

### Miscellaneous Exercise 5 | Q 1.4 | Page 79

Find the slope of the line passing through the following point: (2, - 5), (3, - 1)

#### SOLUTION

Let P = (2, - 5) = (x<sub>1</sub>, y<sub>1</sub>) and Q = (3, - 1) = (x<sub>2</sub>, y<sub>2</sub>) say.

$$\text{Slope of line PQ} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-1 - (-5)}{3 - 2}$$

$$= \frac{-1 + 5}{1}$$

$$= 4.$$

### Miscellaneous Exercise 5 | Q 2.1 | Page 79

Find the slope of the line which makes an angle of 120° with the positive X-axis.

#### SOLUTION

$$\theta = 120^\circ$$

$$\text{Slope of the line} = \tan 120^\circ$$

$$= \tan (180 - 60^\circ)$$

$$= -\tan 60^\circ \quad \dots [\tan(180^\circ - \theta) = -\tan \theta]$$

$$= -\sqrt{3}.$$

### Miscellaneous Exercise 5 | Q 2.2 | Page 79

Find the slope of the line which makes intercepts 3 and - 4 on the axes.

#### SOLUTION

Given, x-intercept of line is 3

and y-intercept of line is - 4

∴ The line intersects X-axis at (3, 0) and Y-axis at (0, - 4).

∴ The line passes through (3, 0) = (x<sub>1</sub>, y<sub>1</sub>) and (0, - 4) = (x<sub>2</sub>, y<sub>2</sub>) say.

$$\begin{aligned}
 \therefore \text{Slope of line} &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{-4 - 0}{0 - 3} \\
 &= \frac{-4}{-3} \\
 &= \frac{4}{3}.
 \end{aligned}$$

### Miscellaneous Exercise 5 | Q 2.3 | Page 79

Find the slope of the line which passes through the points A(−2, 1) and the origin.

#### **SOLUTION**

Required line passes through O(0, 0) = (x<sub>1</sub>, y<sub>1</sub>) and A(−2, 1) = (x<sub>2</sub>, y<sub>1</sub>) say.

$$\begin{aligned}
 \text{Slope of line OA} &= \frac{y_2 - y_1}{x_2 - x_1} \\
 &= \frac{1 - 0}{-2 - 0} \\
 &= \frac{1}{-2} \\
 &= \frac{-1}{2}.
 \end{aligned}$$

### Miscellaneous Exercise 5 | Q 3.1 | Page 79

Find the value of k: if the slope of the line passing through the points (3, 4), (5, k) is 9.

#### **SOLUTION**

Let P(3, 4), Q(5, k).

Slope of PQ = 9 ...[Given]

$$\therefore \frac{k - 4}{5 - 3} = 9$$

$$\therefore \frac{k - 4}{2} = 9$$

$$\therefore k - 4 = 18$$

$$\therefore k = 22.$$

**Miscellaneous Exercise 5 | Q 3.2 | Page 79**

Find the value of k: the points (1, 3), (4, 1), (3, k) are collinear.

**SOLUTION**

The points A(1, 3), B(4, 1) and C(3, k) are collinear.

∴ Slope of AB = Slope of BC

$$\therefore \frac{1 - 3}{4 - 1} = \frac{k - 1}{3 - 4}$$

$$\therefore \frac{-2}{3} = \frac{k - 1}{-1}$$

$$\therefore 2 = 3k - 3$$

$$\therefore k = \frac{5}{3}.$$

**Miscellaneous Exercise 5 | Q 3.3 | Page 79**

Find the value of k: the point P(1, k) lies on the line passing through the points A(2, 2) and B(3, 3).

**SOLUTION**

Given, point P(1, k) lies on the line joining A(2, 2) and B(3, 3).

∴ Slope of AB = Slope of BP

$$\therefore \frac{3 - 2}{3 - 2} = \frac{3 - k}{3 - 1}$$

$$\therefore 1 = \frac{3 - k}{2}$$

$$\therefore 2 = 3 - k$$

$$\therefore k = 1.$$

**Miscellaneous Exercise 5 | Q 4 | Page 79**

Reduce the equation  $6x + 3y + 8 = 0$  into slope-intercept form. Hence, find its slope.

**SOLUTION**

Given equation is  $6x + 3y + 8 = 0$ , which can be written as

$$3y = -6x - 8$$

$$\therefore y = \frac{-6x}{3} - \frac{8}{3}$$

$$\therefore y = -2x - \frac{8}{3}$$

This is of the form  $y = mx + c$  with  $m = -2$

$\therefore y = -2x - \frac{8}{3}$  is in slope-intercept form with slope  $= -2$ .

### Miscellaneous Exercise 5 | Q 5 | Page 79

Verify that A(2, 7) is not a point on the line  $x + 2y + 2 = 0$ .

#### **SOLUTION**

Given equation is  $x + 2y + 2 = 0$ .

Substituting  $x = 2$  and  $y = 7$  in L.H.S. of given equation, we get

$$\text{L.H.S.} = x + 2y + 2$$

$$= 2 + 2(7) + 2$$

$$= 2 + 14 + 2$$

$$= 18$$

$\neq \text{R.H.S.}$

$\therefore$  Point A does not lie on the given line.

### Miscellaneous Exercise 5 | Q 6 | Page 79

Find the X-intercept of the line  $x + 2y - 1 = 0$

#### **SOLUTION**

Given equation of the line is  $x + 2y - 1 = 0$

To find the x-intercept, put  $y = 0$  in given equation of the line

$$\therefore x + 2(0) - 1 = 0$$

$$\therefore x + 0 - 1 = 0$$

$$\therefore x = 1$$

$\therefore$  X-intercept of the given line is 1.

#### **Alternative method:**

Given equation of the line is

$$x + 2y - 1 = 0$$

$$\text{i.e. } x + 2y = 1$$

$$\therefore \frac{x}{1} + \frac{y}{\frac{1}{2}} = 1$$

Comparing with  $\frac{x}{a} + \frac{y}{b} = 1$ , we get  $a = 1$

$\therefore$  X-intercept of the line is 1.

### Miscellaneous Exercise 5 | Q 7 | Page 79

Find the slope of the line  $y - x + 3 = 0$ .

#### SOLUTION

Equation of given line is  $y - x + 3 = 0$

i.e.  $y = x - 3$

Comparing with  $y = mx + c$ , we get

$m = \text{Slope} = 1$ .

### Miscellaneous Exercise 5 | Q 8 | Page 79

Does point A(2, 3) lie on the line  $3x + 2y - 6 = 0$ ? Give reason.

#### SOLUTION

Given equation is  $3x + 2y - 6 = 0$ .

Substituting  $x = 2$  and  $y = 3$  in L.H.S. of given equation, we get

L.H.S. =  $3x + 2y - 6$

=  $3(2) + 2(3) - 6$

= 6

≠ R.H.S.

∴ Point A does not lie on the given line.

### Miscellaneous Exercise 5 | Q 9 | Page 79

Which of the following lines passes through the origin?

1.  $x = 2$
2.  $y = 3$
3.  $y = x + 2$
4.  $2x - y = 0$

#### SOLUTION

Any line passing through origin is of the form  $y = mx$  or  $ax + by = 0$ .

Here in the given option,  $2x - y = 0$  is in the form  $ax + by = 0$ .

### Miscellaneous Exercise 5 | Q 10.1 | Page 79

Obtain the equation of the line which is: parallel to the X-axis and 3 units below it.

#### SOLUTION

Equation of a line parallel to X-axis is  $y = k$ .

Since, the line is at a distance of 3 units below X-axis.

∴  $k = -3$

∴ the equation of the required line is  $y = -3$  i.e.,  $y + 3 = 0$ .

### Miscellaneous Exercise 5 | Q 10.2 | Page 79

Obtain the equation of the line which is: Obtain the equation of the line which is:

**SOLUTION**

Equation of a line parallel to Y-axis is  $x = h$ .

Since, the line is at a distance of 2 units to the left of Y-axis.

$$\therefore h = -2$$

$\therefore$  the equation of the required line is  $x = -2$  i.e.,  $x + 2 = 0$ .

**Miscellaneous Exercise 5 | Q 10.3 | Page 79**

Obtain the equation of the line which is: parallel to the X-axis and making an intercept of 5 on the Y-axis.

**SOLUTION**

Equation of a line parallel to X-axis with y-intercept 'k' is  $y = k$ .

Here, y-intercept = 5

$\therefore$  the equation of the required line is  $y = 5$ .

**Miscellaneous Exercise 5 | Q 10.4 | Page 79**

Obtain the equation of the line which is: parallel to the Y-axis and making an intercept of 3 on the X-axis.

**SOLUTION**

Equation of a line parallel to Y-axis with x-intercept 'h' is  $x = h$ .

Here, x-intercept = 3

$\therefore$  the equation of the required line is  $x = 3$ .

**Miscellaneous Exercise 5 | Q 11.1 | Page 79**

Obtain the equation of the line containing the point: (2, 3) and parallel to the X-axis.

**SOLUTION**

Equation of a line parallel to X-axis is of the form  $y = k$ .

Since, the line passes through (2, 3).

$$\therefore k = 3$$

$\therefore$  the equation of the required line is  $y = 3$ .

**Miscellaneous Exercise 5 | Q 11.2 | Page 79**

Obtain the equation of the line containing the point: (2, 4) and perpendicular to the Y-axis.

**SOLUTION**

Equation of a line perpendicular to Y-axis i.e., parallel to X-axis, is of the form  $y = k$ .

Since, the line passes through (2, 4).

$$\therefore k = 4$$

$\therefore$  the equation of the required line is  $y = 4$ .

**Miscellaneous Exercise 5 | Q 11.3 | Page 79**

Obtain the equation of the line containing the point: (2, 5) and perpendicular to the X-axis.

#### **SOLUTION**

Equation of a line perpendicular to X-axis  
i.e., parallel to Y-axis, is of the form  $x = h$ .

$$\therefore h = 2$$

$\therefore$  the equation of the required line is  $x = 2$ .

#### **Miscellaneous Exercise 5 | Q 12.1 | Page 79**

Find the equation of the line: having slope 5 and containing point A(− 1, 2).

#### **SOLUTION**

Given, slope( $m$ ) = 5 and the line passes through A(− 1, 2).

Equation of the line in slope point form is  $y - y_1 = m(x - x_1)$

$\therefore$  the equation of the required line is

$$y - 2 = 5(x + 1)$$

$$\therefore y - 2 = 5x + 5$$

$$\therefore 5x - y + 7 = 0.$$

#### **Miscellaneous Exercise 5 | Q 12.2 | Page 79**

Find the equation of the line: containing the point (2, 1) and having slope 13.

#### **SOLUTION**

Given, slope( $m$ ) = 13 and the line passes through (2, 1).

Equation of the line in slope point form is

$$y - y_1 = m(x - x_1)$$

$\therefore$  the equation of the required line is

$$y - 1 = 13(x - 2)$$

$$\therefore y - 1 = 13x - 26$$

$$\therefore 13x - y = 25.$$

#### **Miscellaneous Exercise 5 | Q 12.3 | Page 79**

Find the equation of the line: containing the point T(7, 3) and having inclination  $90^\circ$ .

#### **SOLUTION**

Given, Inclination of line =  $\theta = 90^\circ$

$\therefore$  the required line is parallel to Y-axis (or lies on the Y-axis.)

Equation of a line parallel to Y-axis is of the form  $x = h$ .

Since, the line passes through (7, 3).

$$\therefore h = 7$$

$\therefore$  the equation of the required line is  $x = 7$ .

**Miscellaneous Exercise 5 | Q 12.4 | Page 79**

Find the equation of the line: containing the origin and having inclination  $90^\circ$ .

**SOLUTION**

Given, Inclination of line  $= \theta = 90^\circ$

$\therefore$  the required line is parallel to Y-axis (or lies on the Y-axis.)

Equation of a line parallel to Y-axis is of the form  $x = h$ .

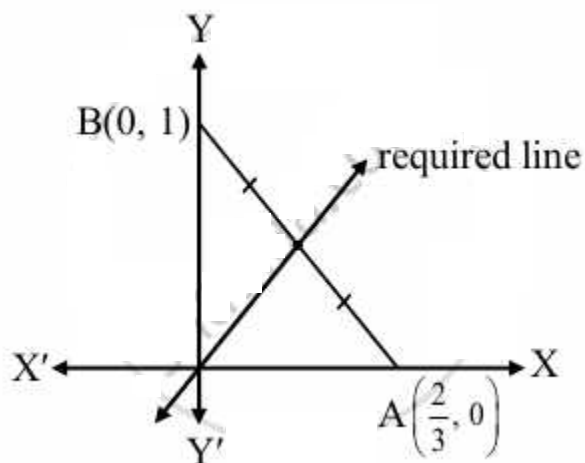
Since, the line passes through origin  $(0, 0)$ .

$\therefore h = 0$

$\therefore$  the equation of the required line is  $x = 0$ .

**Miscellaneous Exercise 5 | Q 12.5 | Page 79**

Find the equation of the line: through the origin which bisects the portion of the line  $3x + 2y = 2$  intercepted between the co-ordinate axes.

**SOLUTION**

Given equation of the line is  $3x + 2y = 2$ .

$$\therefore \frac{3x}{2} + \frac{2y}{2} = 1$$

$$\therefore \frac{x}{\frac{2}{3}} + \frac{y}{1} = 1$$

This equation is of the form  $\frac{x}{a} + \frac{y}{b} = 1$ , with  $a = \frac{2}{3}$ ,  $b = 1$ .

∴ the line  $3x + 2y = 2$  intersects the X-axis at  $A\left(\frac{2}{3}, 0\right)$  and Y-axis at  $B(0, 1)$ .

Required line is passing through the midpoint of AB.

$$\therefore \text{Midpoint of AB} = \left( \frac{\frac{2}{3} + 0}{2}, \frac{0 + 1}{2} \right) = \left( \frac{1}{3}, \frac{1}{2} \right)$$

∴ Required line passes through  $(0, 0)$  and  $\left(\frac{1}{3}, \frac{1}{2}\right)$ .

Equation of the line in two point form is

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

∴ the equation of the required line is

$$\frac{y - 0}{\frac{1}{2} - 0} = \frac{x - 0}{\frac{1}{3} - 0}$$

$$\therefore 2y = 3x$$

$$\therefore 3x - 2y = 0.$$

#### Miscellaneous Exercise 5 | Q 13 | Page 80

Find the equation of the line passing through the points  $A(-3, 0)$  and  $B(0, 4)$ .

#### **SOLUTION**

Since, the required line passes through the points  $A(-3, 0)$  and  $B(0, 4)$ .

Equation of the line in two point form is

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

Here,  $(x_1, y_1) = (-3, 0)$  and  $(x_2, y_2) = (0, 4)$

∴ the equation of the required line is

$$\frac{y - 0}{4 - 0} = \frac{x - (-3)}{0 - (-3)}$$

$$\therefore \frac{y}{4} = \frac{x+3}{3}$$

$$\therefore 4x + 12 = 3y$$

$$\therefore 4x - 3y + 12 = 0.$$

### Miscellaneous Exercise 5 | Q 14.1 | Page 80

Find the equation of the line: having slope 5 and making intercept 5 on the X-axis.

#### **SOLUTION**

Since, the x-intercept of the required line is 5.

$\therefore$  it passes through (5, 0).

Also, slope(m) of the line is 5

Equation of the line in slope point form is

$$y - y_1 = m(x - x_1)$$

$\therefore$  the equation of the required line is

$$y - 0 = 5(x - 5)$$

$$\therefore y = 5x - 25$$

$$\therefore 5x - y - 25 = 0.$$

### Miscellaneous Exercise 5 | Q 14.2 | Page 80

Find the equation of the line: having an inclination  $60^\circ$  and making intercept 4 on the Y-axis.

#### **SOLUTION**

Given, Inclination of line =  $\theta = 60^\circ$

$\therefore$  slope of the line (m) =  $\tan \theta = \tan 60^\circ = \sqrt{3}$  and the y-intercept of the required line is 4.

$\therefore$  it passes through (0, 4).

Equation of the line in slope point form is

$$y - y_1 = m(x - x_1)$$

$\therefore$  the equation of the required line is

$$y - 4 = \sqrt{3}(x - 0)$$

$$\therefore y - 4 = \sqrt{3}x$$

$$\therefore \sqrt{3}x - y + 4 = 0.$$

**Miscellaneous Exercise 5 | Q 15.1 | Page 80**

The vertices of a triangle are A (1, 4), B (2, 3) and C (1, 6). Find equations of the sides

**SOLUTION**

Vertices of  $\triangle ABC$  are A(1, 4), B(2, 3) and C(1, 6)

Equation of the line in two point form is

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

Equation of side AB is

$$\frac{y - 4}{3 - 4} = \frac{x - 1}{2 - 1}$$

$$\therefore \frac{y - 4}{-1} = \frac{x - 1}{1}$$

$$\therefore y - 4 = -1(x - 1)$$

$$\therefore x + y = 5$$

Equation of side BC is

$$\frac{y - 3}{6 - 3} = \frac{x - 2}{1 - 2}$$

$$\therefore \frac{y - 3}{3} = \frac{x - 2}{-1}$$

$$\therefore -1(y - 3) = 3(x - 2)$$

$$\therefore 3x + y = 9$$

Since, both the points A and C have same x co-ordinates i.e. 1

$\therefore$  the points A and C lie on a line parallel to Y-axis.

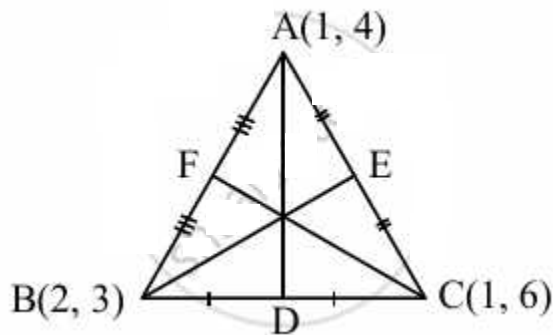
$\therefore$  the equation of side AC is  $x = 1$ .

**Miscellaneous Exercise 5 | Q 15.2 | Page 80**

The vertices of a triangle are A (1, 4), B (2, 3) and C (1, 6). Find equations of the medians

**SOLUTION**

Vertices of  $\triangle ABC$  are  $A(1, 4)$ ,  $B(2, 3)$  and  $C(1, 6)$   
 Let  $D$ ,  $E$  and  $F$  be the midpoints of sides  $BC$ ,  
 $AC$  and  $AB$  respectively of  $\triangle ABC$ .



$$\text{Then } D = \left( \frac{2+1}{2}, \frac{3+6}{2} \right) = \left( \frac{3}{2}, \frac{9}{2} \right)$$

$$E = \left( \frac{1+1}{2}, \frac{6+4}{2} \right) = (1, 5)$$

$$F = \left( \frac{1+2}{2}, \frac{4+3}{2} \right) = \left( \frac{3}{2}, \frac{7}{2} \right)$$

Equation of median  $AD$  is

$$\frac{y-4}{\frac{9}{2}-4} = \frac{x-1}{\frac{3}{2}-1}$$

$$\therefore \frac{y-4}{\frac{1}{2}} = \frac{x-1}{\frac{1}{2}}$$

$$\therefore y-4 = x-1$$

$$\therefore x-y+3=0$$

Equation of median  $BE$  is

$$\frac{y-3}{5-3} = \frac{x-2}{1-2}$$

$$\therefore \frac{y-3}{2} = \frac{x-2}{-1}$$

$$\therefore -1(y + 3) = 2(x - 2)$$

$$\therefore -y + 3 = 2x - 4$$

$$\therefore 2x + y = 7$$

Equation of median CF is

$$\frac{y - 6}{\frac{7}{2} - 6} = \frac{x - 1}{\frac{3}{2} - 1}$$

$$\therefore \frac{y - 6}{-\frac{5}{2}} = \frac{x - 1}{\frac{1}{2}}$$

$$\therefore \frac{y - 6}{-5} = \frac{x - 1}{1}$$

$$\therefore y - 6 = -5(x - 1)$$

$$\therefore y - 6 = -5 + 5$$

$$\therefore 5x + y - 11 = 0.$$

### Miscellaneous Exercise 5 | Q 15.3 | Page 80

The vertices of a triangle are A (1, 4), B (2, 3) and C (1, 6). Find equations of Perpendicular bisectors of sides

#### **SOLUTION**

Vertices of  $\triangle ABC$  are A(1, 4), B(2, 3) and C(1, 6)

$\therefore$  Slope of perpendicular bisector of BC is  $\frac{1}{3}$  and the line passes through  $\left(\frac{3}{2}, \frac{9}{2}\right)$ .

$\therefore$  Equation of the perpendicular bisector of side BC is

$$\left(y - \frac{9}{2}\right) = \frac{1}{3} \left(x - \frac{3}{2}\right)$$

$$\therefore \frac{2y - 9}{2} = \frac{1}{3} \left(\frac{2x - 3}{2}\right)$$

$$\therefore 3(2y - 9) = (2x - 3)$$

$$\therefore 2x - 6y + 24 = 0$$

$$\therefore x - 3y + 12 = 0$$

Since, both the points A and C have same x co-ordinates i.e. 1

∴ the points A and C lie on the line  $x = 1$ .

AC is parallel to Y-axis and therefore, perpendicular bisector of side AC is parallel to X-axis.

Since, the perpendicular bisector of side AC passes through E(1, 5).

∴ the equation of perpendicular bisector of side AC is  $y = 5$ .

$$\text{Slope of side AB} = \left( \frac{3 - 4}{2 - 1} \right) = -1$$

∴ Slope of perpendicular bisector of AB is 1 and the line passes through  $\left( \frac{3}{2}, \frac{7}{2} \right)$ .

∴ Equation of the perpendicular bisector of side AB is

$$\left( y - \frac{7}{2} \right) = 1 \left( x - \frac{3}{2} \right)$$

$$\therefore \frac{2y - 7}{2} = \frac{2x - 3}{2}$$

$$\therefore 2y - 7 = 2x - 3$$

$$\therefore 2x - 2y + 4 = 0$$

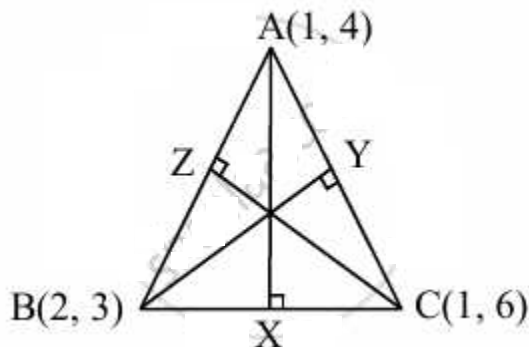
$$\therefore x - y + 2 = 0.$$

### Miscellaneous Exercise 5 | Q 15.4 | Page 80

The vertices of a triangle are A (1, 4), B (2, 3) and C (1, 6). Find equations of altitudes of  $\triangle ABC$

#### **SOLUTION**

Vertices of  $\triangle ABC$  are A(1, 4), B(2, 3) and C(1, 6)



Let AX, BY and CZ be the altitudes through the vertices A, B and C respectively of  $\triangle ABC$ .

Slope of BC = -3

$$\therefore \text{slope of } AX = \frac{1}{3} \quad \dots[\because AX \perp BC]$$

Since, altitude AX passes through (1, 4) and has slope  $\frac{1}{3}$

$\therefore$  equation of altitude AX is

$$y - 4 = \frac{1}{3}(x - 1)$$

$$\therefore 3y - 12 = x - 1$$

$$\therefore x - 3y + 11 = 0$$

Since, both the points A and C have same x co-ordinates i.e. 1  
 $\therefore$  the points A and C lie on the line  $x = 1$ .

AC is parallel to Y-axis and therefore, altitude  
BY is parallel to X-axis.

Since, the altitude BY passes through B(2, 3).

$\therefore$  the equation of altitude BY is  $y = 3$ .

Also, slope of AB = - 1

$\therefore$  slope of CZ = 1  $\dots[\because CZ \perp AB]$

Since, altitude CZ passes through (1, 6) and has slope 1

$\therefore$  equation of altitude CZ is

$$\therefore y - 6 = 1(x - 1)$$

$$\therefore x - y + 5 = 0.$$